



Instructor: Date:

Learner: Regional Branch:

Program: Center:

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Let's Start!

Let's see what software and hardware engineers do.



Learning Outcome:

Identify key information about people's jobs and places of work in spoken or written contexts.

Ask your partner:

- What type of job related to computing would you like to have?
- Would you rather work with hardware or software? Why?

Can Do Statements:

- I can talk about duties and responsibilities related to job positions in IT.
- I can give detailed information related to software and hardware engineering in info systems.
- I can compare both the software engineer and the hardware engineer.



Let's Explore!

Katy and Randy work as engineers.



Instructions: Read the text below.

Kate: Excuse me, are you Randy Marshall?

Randy: Yeah! I'm Randy. Who are you?

Kate: Randy, don't you remember me? You make me feel like a cog in the machine!

Randy: Sorry, I don't.

Kate: I'm Kate Johnson, your university classmate.

Randy: Oh, yes! My apologies, Kate. Please, don't blow a fuse! How's everything going?

Kate: Well, I work as a hardware engineer. What about you? What do you do?

Randy: Pretty much the same as you. I spend my days in front of a computer. But, in my case, I work as a software engineer.

Kate: Really? What do you have to do in that field?

Randy: Well, I have to design particular programs, I research new applications, and I handle support all the time. In your case, what do you got to do?

Kate: I have to check the hardware configuration of computers. I upgrade layouts and components to make computers work faster and in a more efficient way.

Randy: Hum, I see. So, hardware engineers are more functional than software engineers in the sense that you need to know how computers operate.

Kate: Exactly! And software engineers need to be more creative than hardware engineers because you make programs.

Randy: That's it! Moreover, I think technical responsibilities are more important for you than programming responsibilities.

Kate: Yes! And, you are more skillful in research than us.

Randy: You know, I make these comparisons because I want to work as an IT specialist. So, I need to be competitive in both fields; hardware and software.

Kate: You're right! In addition, main duties and responsibilities of that job are related to networks and database management. Also, you have to assess the practicality of technology systems and, you need to keep up to date on technology advancements.



**Let's Practice!**

Kate and Randy mention some of their duties.

Instructions: Select the correct option to correctly complete each sentence.

1. These engineers (manage / manages / managed) networks to avoid system issues in the company.
2. An engineer (evaluate / evaluated / evaluates) the convenience of the technology networks.
3. Updating the system (represents / represent / to represent) one of the main duties of software and hardware engineers.

**Let's Practice!**

John compares software and hardware engineers.

Instructions: Circle the correct answer.

1. Hardware engineers are _____ in how the computer operation works than software engineers.
a. more specialized
b. most specialized
c. specialized
2. Software engineers have _____ chances to take various short term and long term certification courses than hardware engineers.
a. big
b. the biggest
c. bigger
3. Hardware engineers are _____ software engineers for the company. They aren't a cog in the machine.
a. important
b. as important as
c. more important
4. An IT security manager is in charge of setting _____ practices for securing hardware, backups and VPNs.
a. better
b. most good
c. more good
5. A person in charge of managing the relational databases and their access rights _____ a DBA.
a. were
b. is
c. are





Extension Activity

Hardware and software engineers in your community.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Find someone who can talk to you about hardware and software engineers.
- Create a chart comparing their main characteristics.
- Share your chart with the class.

Self-Assessment

Yes

No

- I look for someone who can talk to me about hardware and software engineers.
- I create a comparative chart describing main characteristics.
- I share my work with my class.

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Target Words

- **hardware** (noun): components of a computer, such as the hard disk drive, the monitor, the keyboard and others
e.g.
- **research** (verb): to investigate
e.g.
- **support** (noun): a service assistance and supervision
e.g.
- **skillful** (adjective): experienced and competent
e.g.
- **database** (noun): a system that contains information
e.g.
- **keep up to date** (expression): to be informed all the time about something
e.g.



Learning Points

Simple Present Tense Review.

You can use the **simple present** to talk about duties and occupations.

To make affirmative statements in **Simple Present**, you just need to follow the next structure:

Subject + **Verb** + Complement

Example:

*They **work** as engineers in a software company.*

Remember that If the subject pronoun is **he**, **she** or **it**, you must add the letter **s** at the end of the verb.

Example:

*Sam **works** as an engineer in a software company.*

In some cases, you will need to add **-es** or **-ies** to the verb. Take a look at the following chart:

Rule	Examples
Most verbs just need s .	design - designs produce - produces train - trains
Add es to verbs that end in: <i>ch</i> , <i>sh</i> , <i>o</i> and <i>x</i> .	watch - watches wash - washes go - goes fix - fixes
Add ies to verbs that end in a <i>consonant + y</i> .	study - studies cry - cries carry - carries However,
Add s if the verb ends in a <i>vowel + y</i> .	play - plays buy - buys say - says



Learning Points

Idioms about Technology.

We can find different expressions and idioms that contain vocabulary related to technology.

Remember that we shouldn't deduce the meaning of an idiom by the meaning of the word in it. Instead, we need to memorize what it really means. Let's take a look at some examples of idioms:

To blow a fuse means to become really angry about something.

Example: *Harry, **don't blow a fuse**. I didn't mean to break your mouse.*

Cog in the machine refers to someone with little importance in a big organization.

Example: *I don't want to be just **a cog in the machine**. I want to have more responsibilities.*

To hit the panic button means to lose control and panic.

Example: *We don't know how to fix this software. But we are not going **to hit the panic button** yet.*